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React ES6 Destructuring

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Destructuring in React

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Destructuring is a JavaScript feature that allows you to extract values from objects or arrays into distinct variables. In React, it's commonly used with props, hooks, and state management.

Note: Destructuring makes React code cleaner and more readable by reducing repetitive object and array access.

Destructuring Arrays

Here is the old way of assigning array items to a variable:

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```
const vehicles = ['mustang', 'f-150', 'expedition'];

// old way
const car = vehicles[0];
const truck = vehicles[1];
const suv = vehicles[2];

//You can now access each variable separately:
document.getElementById('demo').innerHTML = truck;
```

Try it Yourself »

Here is the new way of assigning array items to a variable:

Example

With destructuring:

```
const vehicles = ['mustang', 'f-150', 'expedition'];

const [car, truck, suv] = vehicles;

//You can now access each variable separately:
document.getElementById('demo').innerHTML = truck;
```

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When destructuring *arrays*, the order that variables are declared is important.

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```
const vehicles = ['mustang', 'f-150', 'expedition'];

const [car,, suv] = vehicles;
```

Destructuring comes in handy when a function returns an array:

Example

```
function dateInfo(dat) {
  const d = dat.getDate();
  const m = dat.getMonth() + 1;
  const y = dat.getFullYear();

  return [d, m, y];
}

const [date, month, year] = dateInfo(new Date());
```

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Destructuring Objects

You can use destructuring to extract the values from an object:

Example

Unpack the values from an object:

```
const person = {
  firstName: "John",
  lastName: "Doe",
  age: 50
};

// Destructuring
let {firstName, lastName, age} = person;

//You can now access each variable separately:
document.getElementById("demo").innerHTML = firstName;
```

Try it Yourself »

For *objects*, the *order* of the properties does not matter:

Example

Unpack the values in a random order:

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```
  age: 50
};

// Destructuring
let {lastName, age, firstName} = person;
```

Try it Yourself »

You can extract only the value(s) you want:

Example

Extract only firstName from the object:

```
const person = {
  firstName: "John",
  lastName: "Doe",
  age: 50
};

// Destructuring
let {firstName} = person;
```

Try it Yourself »

For potentially missing properties we can set default values:

Example

Set a default value for the missing property:

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```
  age: 50
};

// Destructuring
let {firstName, lastName, age, country = "Norway"} = person;
```

Try it Yourself »

We can also destructure deeply nested objects by referencing the nested object then using a colon and curly braces to again destructure the items needed from the nested object:

Example

```
const person = {
  firstName: "John",
  lastName: "Doe",
  age: 50,
  car: {
    brand: 'Ford',
    model: 'Mustang',
  }
};

// Destructuring
let {firstName, car: { brand, model }} = person;

let message = `My name is ${firstName}, and I drive a ${brand} ${model}.`;
```

Try it Yourself »

Read more about Destructuring in our JavaScript Tutorial.

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Destructuring is particularly useful in React for working with props, hooks, and API responses. It helps make your code more concise and easier to read.



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Props Destructuring

When a component receives props, you can use destructuring to extract the values you need.

Check out the difference between using and not using destructuring:

```
//Using destructuring:
function Greeting({ name, age }) {
  return <h1>Hello, {name}! You are {age} years old.</h1>;
}
```

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```
  return <h1>Hello, {props.name}! You are {props.age} years old.</h1>;
}
```

Let's see a working example:

Example

Using destructuring to extract props:

```
import { createRoot } from 'react-dom/client'

function Greeting({ name, age }) {
  return <h1>Hello, {name}! You are {age} years old.</h1>;
}

createRoot(document.getElementById('root')).render(
  <Greeting name="John" age={25} />
);
```

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useState Hook Destructuring

When a component uses the *useState* hook, we use destructuring to extract the values from it.

Example

Using destructuring to extract values from useState:

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```
  // Destructuring the array returned by useState
  const [count, setCount] = useState(0);

  return (
    <button onClick={() => setCount(count + 1)}>
      Count: {count}
    </button>
  );
}

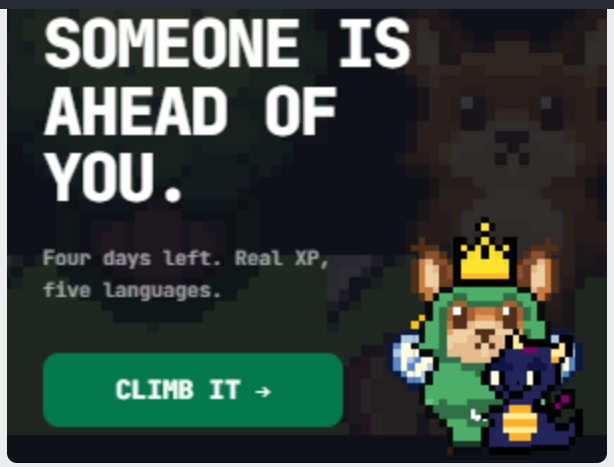
createRoot(document.getElementById('root')).render(
  <Counter />
);
```

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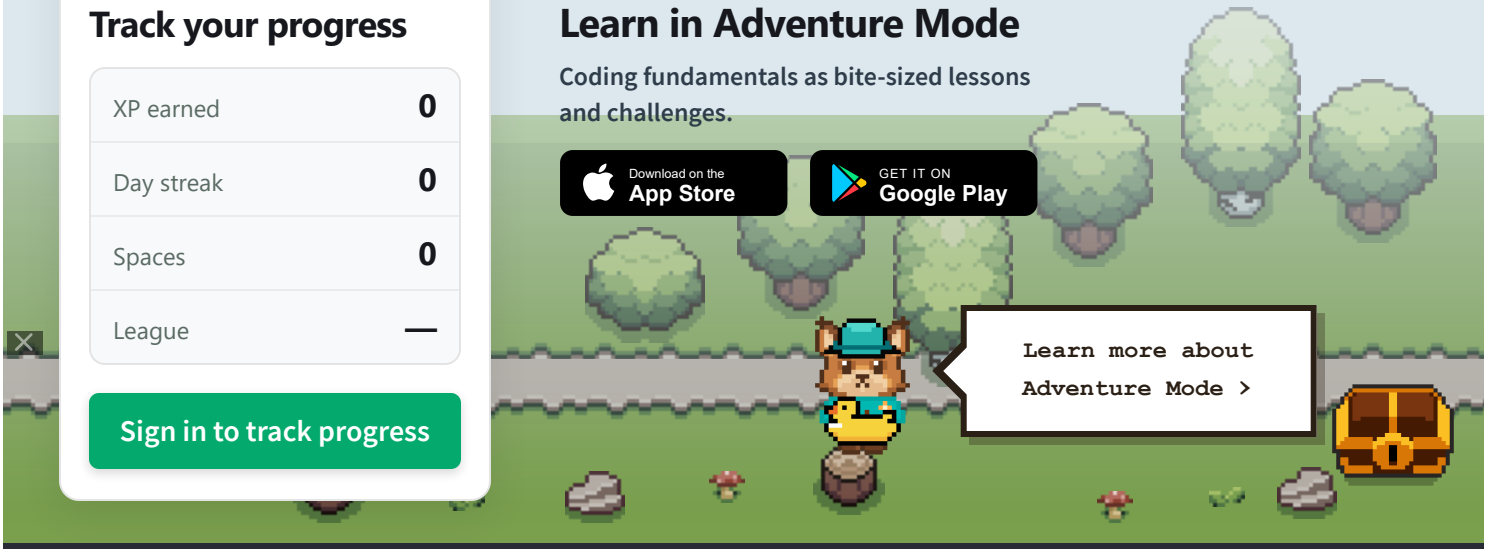
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